

CHAPTER 24

HELISPECTRAL FIELD

Unified Harmonic Spectral Geometry, Coherence and Propagation

Simon Ellis - Universal Harmonic Theory - Conceptual Mathematical Draft

Scientific status. The wave, spectral, Fourier, coherence and scalar-field equations used below are established mathematical tools. “Helispectral Field”, ZNE, ZNY and the C-G universal interpretation are proposed elements of the Universal Harmonic Theory and remain hypotheses until their variables, dimensions, coupling laws, boundary conditions and distinct experimental predictions are fully specified.

24.1 Definition and Field Content

The Helispectral Field is defined as a coherent distribution of propagating modes in which amplitude, frequency, wavelength, phase, polarisation and geometric structure are represented within one spectral field. Its simplest scalar form is

$$\mathcal{H}_s(x,t) = \Phi(x,t) Z_{NE} Z_{NY} B_{CG} \quad (24.1)$$

Here Φ is the underlying propagating field; Z_{NE} and Z_{NY} are dimensionless conceptual balance factors; and B_{CG} is the C-G harmonic-balance factor. For dimensional consistency, all three modifiers must be dimensionless unless they are replaced by explicitly dimensional couplings.

24.2 Spectral Decomposition

A real one-dimensional Helispectral Field can be expanded into discrete modes:

$$\mathcal{H}_s(x,t) = \sum_n A_n \cos(k_n x - \omega_n t + \varphi_n) \quad (24.2)$$

For a continuous spectrum,

$$\mathcal{H}_s(x,t) = \text{Re}\{ \int_0^\infty \tilde{A}(k) \exp[i(kx - \omega(k)t)] dk \} \quad (24.3)$$

where $\tilde{A}(k)$ is the complex spectral amplitude and $\omega(k)$ is the dispersion relation. The spectral energy density may be written

$$S_H(k) = |\tilde{A}(k)|^2 \quad (24.4)$$

24.3 Linear Propagation

In an ideal nondispersive medium, the field obeys the standard wave equation:

$$\partial^2 \mathcal{H}_s / \partial t^2 - c_H^2 \nabla^2 \mathcal{H}_s = 0 \quad (24.5)$$

A monochromatic solution is

$$\mathcal{H}_s(x,t) = A_H \cos(kx - \omega t + \varphi_H), \quad \omega = c_H k \quad (24.6)$$

The group and phase velocities are

$$v_p = \omega/k, \quad v_g = d\omega/dk \quad (24.7)$$

24.4 Damping, Dispersion and Scatter-Field Propagation

A more general phenomenological propagation equation is

$$\partial^2 \mathcal{H}_s / \partial t^2 + 2\gamma \partial \mathcal{H}_s / \partial t - c_H^2 \nabla^2 \mathcal{H}_s + d \nabla^4 \mathcal{H}_s + V_{sc}(r) \mathcal{H}_s = J_H(r,t) \quad (24.8)$$

where γ is damping, d controls higher-order dispersion, V_{sc} is an effective scattering potential and J_H is a source. The total field is decomposed as

$$\mathcal{H}_{tot} = \mathcal{H}_{inc} + \sum_j \mathcal{H}_{sc,j} \quad (24.9)$$

and the measurable intensity is

$$I_H(r,t) \propto |\mathcal{H}_{tot}(r,t)|^2 \quad (24.10)$$

$$I_H = |\mathcal{H}_{inc}|^2 + |\mathcal{H}_{sc}|^2 + 2 \operatorname{Re}(\mathcal{H}_{inc} \mathcal{H}_{sc}^*). \quad (24.11)$$

24.5 C-G Perfect-Fifth Balance

Represent the C and G sectors by two modes:

$$\mathcal{H}_C = A_C \cos(k_C x - \omega_C t + \varphi_C) \quad (24.12)$$

$$\mathcal{H}_G = A_G \cos(k_G x - \omega_G t + \varphi_G) \quad (24.13)$$

$$\mathcal{H}_{CG} = \mathcal{H}_C + \mathcal{H}_G. \quad (24.14)$$

The ideal just-intonation perfect-fifth condition is

$$\omega_G / \omega_C = f_G / f_C = 3/2. \quad (24.15)$$

Because a 3:2 relation is periodic rather than phase-identical, a suitable locking phase is

$$\Delta\Phi_{2:3} = 2\varphi_G - 3\varphi_C \quad (24.16)$$

$$\Gamma_{2:3} = |\langle \exp(i\Delta\Phi_{2:3}) \rangle|, \quad 0 \leq \Gamma_{2:3} \leq 1. \quad (24.17)$$

Ideal C-G locking corresponds to $\Gamma_{2:3} = 1$; damping, detuning and noise generally produce $\Gamma_{2:3} < 1$.

24.6 Helispectral Balance Operator

A bounded dimensionless C-G balance factor can be defined as

$$B_{CG} = \Gamma_{2:3} \exp[-(\delta_{CG}/\sigma_{CG})^2] \quad (24.18)$$

where

$$\delta_{CG} = 2\omega_G - 3\omega_C \quad (24.19)$$

is the 3:2 detuning and σ_{CG} sets the tolerated spectral width. Then $0 \leq B_{CG} \leq 1$, with $B_{CG} = 1$ only at ideal phase locking and zero detuning.

24.7 ZNE and ZNY as Explicit Modifiers

To avoid confusing the conceptual notation with standard zero-noise extrapolation, define the modifiers operationally. One possible normalised construction is

$$Z_{NE} = \exp(-\eta_N \mathcal{N}), \quad 0 < Z_{NE} \leq 1 \quad (24.20)$$

$$Z_{NY} = \exp(-\eta_Y \mathcal{Y}^2), \quad 0 < Z_{NY} \leq 1 \quad (24.21)$$

where $\mathcal{N}$ is a dimensionless noise or decoherence measure, $\mathcal{Y}$ is a dimensionless displacement from the proposed equilibrium condition, and η_N, η_Y are non-negative sensitivity constants. The effective coherent field becomes

$$\mathcal{H}_{\text{eff}} = Z_{NE} Z_{NY} B_{CG} \mathcal{H}_s. \quad (24.22)$$

24.8 Coherence and Resonance Distillation

A local phase-coherence statistic is

$$\Gamma_H(t) = |(1/N) \sum_n \exp[i\varphi_n(t)]|. \quad (24.23)$$

The coherent portion of a measured field can be extracted by spectral projection:

$$\mathcal{H}_{\text{dist}}(t) = \mathcal{F}^{-1}\{W(\omega - \omega_0) \mathcal{F}[\mathcal{H}_{\text{tot}}(t)]\}. \quad (24.24)$$

A C-G coherence-weighted projection is

$$\mathcal{H}_{CG,\text{dist}} = \Gamma_{2:3} [a_C u_C + a_G u_G], \quad (24.25)$$

$$a_n = \langle u_n, \mathcal{H}_{\text{tot}} \rangle / \langle u_n, u_n \rangle. \quad (24.26)$$

24.9 Nonlinear Helispectral Dynamics

Nonlinear generation and mode coupling may be represented by

$$\partial^2 \mathcal{H}_s / \partial t^2 - c_H^2 \nabla^2 \mathcal{H}_s + 2\gamma \partial \mathcal{H}_s / \partial t + \alpha \mathcal{H}_s + \beta \mathcal{H}_s^2 + \chi \mathcal{H}_s^3 + \kappa_{CG} \mathcal{H}_C \mathcal{H}_G = J_{sc}. \quad (24.27)$$

Quadratic and cubic terms generate new frequencies:

$$\omega_{\text{out}} \in \{0, 2\omega_1, 2\omega_2, \omega_1 + \omega_2, |\omega_1 - \omega_2|, 3\omega_1, 3\omega_2, 2\omega_1 \pm \omega_2, 2\omega_2 \pm \omega_1\}. \quad (24.28)$$

24.10 Lagrangian and Energy Functional

A covariant scalar-field model may begin with the Lagrangian density

$$\mathcal{L}_H = -\frac{1}{2} g^{\mu\nu} (\nabla_\mu \mathcal{H}_s) (\nabla_\nu \mathcal{H}_s) - V(\mathcal{H}_s) + J_H \mathcal{H}_s. \quad (24.29)$$

For

$$V(\mathcal{H}_s) = \frac{1}{2}m_H^2 \mathcal{H}_s^2 + (\lambda_3/3!) \mathcal{H}_s^3 + (\lambda_4/4!) \mathcal{H}_s^4, \quad (24.30)$$

variation gives

$$\square_g \mathcal{H}_s - m_H^2 \mathcal{H}_s - (\lambda_3/2) \mathcal{H}_s^2 - (\lambda_4/6) \mathcal{H}_s^3 = -J_H. \quad (24.31)$$

In flat space, a corresponding energy functional is

$$E_H = \int d^3x [\frac{1}{2}(\partial_t \mathcal{H}_s)^2 + \frac{1}{2}c_H^2 |\nabla \mathcal{H}_s|^2 + V(\mathcal{H}_s)]. \quad (24.32)$$

24.11 Curved Spacetime and Harmonic Geometry

The curved-spacetime d'Alembertian is

$$\square_g \mathcal{H}_s = (1/\sqrt{-g}) \partial_\mu (\sqrt{-g} g^{\mu\nu} \partial_\nu \mathcal{H}_s). \quad (24.33)$$

A conservative scalar-curvature coupling can be introduced as

$$\mathcal{L}_{\text{cpl}} = -\frac{1}{2} \xi_H R \mathcal{H}_s^2. \quad (24.34)$$

The resulting field equation becomes

$$\square_g \mathcal{H}_s - m_H^2 \mathcal{H}_s - \xi_H R \mathcal{H}_s - (\lambda_3/2) \mathcal{H}_s^2 - (\lambda_4/6) \mathcal{H}_s^3 = -J_H. \quad (24.35)$$

The associated stress-energy tensor is obtained by metric variation:

$$T^\wedge(H)_{\{\mu\nu\}} = -(2/\sqrt{-g}) \delta S_H / \delta g^{\{\mu\nu\}}. \quad (24.36)$$

Einstein's equation would then read

$$G_{\{\mu\nu\}} + \Lambda g_{\{\mu\nu\}} = (8\pi G/c^4)(T^\wedge\{\text{matter}\}_{\{\mu\nu\}} + T^\wedge\{H\}_{\{\mu\nu\}}). \quad (24.37)$$

24.12 Compact Master Equation

A compact phenomenological master equation combining propagation, damping, dispersion, nonlinear generation, curvature coupling, C-G balance, ZNE, ZNY and scatter-field forcing is

$$\square_g \mathcal{H}_s + 2\gamma u^\mu \nabla_\mu \mathcal{H}_s + d(\square_g)^2 \mathcal{H}_s - m_H^2 \mathcal{H}_s - \xi_H R \mathcal{H}_s - (\lambda_3/2) \mathcal{H}_s^2 - (\lambda_4/6) \mathcal{H}_s^3 = -J_{\text{sc}} + \kappa_H Z_{\text{NE}} Z_{\text{NY}} B_{\text{CG}} \mathcal{H}_s. \quad (24.38)$$

This is a research ansatz, not an established field equation. Each coefficient must be assigned units and independent operational meaning before numerical or experimental use.

24.13 Stability Conditions

For a homogeneous equilibrium $\mathcal{H}_s = H_0$, equilibrium requires

$$dV_{\text{eff}}/dH|_{(H_0)} = 0, \quad (24.39)$$

and local stability requires

$$d^2V_{\text{eff}}/dH^2|_{(H_0)} > 0. \quad (24.40)$$

For the quartic potential,

$$m_H^2 + \lambda_3 H_0 + (\lambda_4/2) H_0^2 - \xi_{HR} > 0. \quad (24.41)$$

24.14 Testable Quantities

- Harmonic conversion efficiencies $\eta_2 = P(2\omega)/P(\omega)$ and $\eta_3 = P(3\omega)/P(\omega)$.
- The 3:2 phase-locking statistic $\Gamma_{2:3}$ and detuning dependence $B_{CG}(\delta_{CG})$.
- Spectral linewidth, damping rate γ , quality factor $Q = \omega_0/(2\gamma)$, and coherence time τ_c .
- Scattering cross-sections and angular distributions at newly generated frequencies.
- Any measurable deviation between the baseline model $\kappa_H = 0$ and the proposed coupled model $\kappa_H \neq 0$.

24.15 Final Definition

The Helispectral Field is the proposed unified spectral state of a propagating system in which modal amplitude, phase, frequency, geometry, scattering and coherence are represented together. Its C-G sector is defined by a 3:2 periodic-locking condition, while ZNE and ZNY are treated as explicit dimensionless modifiers of coherent field strength. The framework remains conceptual until derived from a complete action with dimensional consistency, well-posed boundary conditions and experimentally distinguishable predictions.